Sai Sony Malapati

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Education

Stony Brook University — Stony Brook, NY
M.S. in Data Science — *Expected May 2027*

Nagarjuna College of Engineering and Technology — Bengaluru, India
B.E. in Computer Science (Data Science), CGPA: 9.3/10 — *2021 – 2025*

Technical Skills

Languages: Java, Python, C++, C, JavaScript, TypeScript, SQL
Frameworks & Libraries: Spring Boot, Spring Security, Spring Data JPA, React, FastAPI, Flask, Node.js, Express.js
Databases & Infrastructure: PostgreSQL, Redis, MySQL, MongoDB, Docker, Docker Compose, Flyway
Engineering Tools: Git, GitHub, Maven, GitHub Actions, OpenAPI/Swagger, Prometheus, k6, Linux
Concepts: REST APIs, JWT Authentication, Caching, Rate Limiting, Database Indexing, CI/CD, Testing, OOP, Data Structures & Algorithms

Experience

Graduate Research Assistant – Stony Brook University — Stony Brook, NY
Python, R, Statistical Modeling, Data Analysis — *2026 – Present*

- Analyzed clinical datasets containing 31,000+ patients using R, performing data cleaning, feature engineering, statistical modeling, and reproducible data analysis.
- Developed reusable analytical workflows to evaluate demographic, clinical, and comorbidity factors across patient populations and research subgroups.
- Applied Cox regression, random forest, and variable-importance analysis to identify predictors and generate research-ready tables, results, and statistical interpretations.

Data Engineer Intern – Arlanto Private Limited — Bengaluru, India
Python, SQL, ETL, Data Processing — *Feb 2025 – Jul 2025*

- Developed Python and SQL ETL workflows to ingest, clean, transform, validate, and integrate structured data for downstream analytics and reporting.
- Implemented automated data-quality and validation checks to identify inconsistencies and improve reliability across data-processing workflows.
- Consolidated data from multiple sources into reusable datasets supporting querying, reporting, and ad-hoc analysis.

MERN Stack Development Intern – BitstreamIo Technologies — Remote
React, Node.js, Express.js, MongoDB — *2024*

- Developed full-stack application features using React, Node.js, Express.js, and MongoDB, integrating frontend components with backend services and database operations.
- Designed and integrated REST API endpoints to support application workflows, data retrieval, and persistent storage.
- Debugged and tested application functionality and collaborated through Git-based software development workflows.

Projects

Scalable Job Application Tracker — *GitHub*
Java, Spring Boot, React, PostgreSQL, Redis, Docker, GitHub Actions

- Designed and built a full-stack job application platform using Java, Spring Boot, React, PostgreSQL, and Redis, implementing JWT access/refresh authentication, refresh-token revocation, user-level authorization, search, filtering, and pagination.
- Optimized PostgreSQL access patterns with composite indexes and Redis caching; validated index usage using `EXPLAIN ANALYZE` on 50K synthetic rows, observing approximately 0.668 ms execution for the indexed local query-plan test.
- Added Redis-backed per-user rate limiting, Flyway migrations, Swagger/OpenAPI, Prometheus/Actuator metrics, request correlation logging, automated tests, Docker Compose, and GitHub Actions full-stack CI.

FloatChat – AI-Powered Ocean Data Analytics Platform — *GitHub*
Python, FastAPI, React, TypeScript, PostgreSQL, ChromaDB, Gemini

- Built a full-stack conversational analytics platform using React/TypeScript, FastAPI, PostgreSQL, and Python for querying and exploring large-scale ARGO oceanographic datasets.
- Developed RESTful backend services and data-ingestion pipelines using FastAPI, SQLAlchemy, Pydantic, pandas,

and xarray to process, validate, and persist NetCDF ocean measurements.
- Implemented a Retrieval-Augmented Generation pipeline with ChromaDB and Gemini for semantic retrieval and natural-language-to-SQL query generation.

Intelligent Assistant *GitHub*
Python, Flask, Gemini API, Search APIs, Image Processing

- Developed a web-based intelligent assistant integrating Gemini API, search functionality, and image-processing workflows for context-aware user interactions.
- Integrated external APIs and backend application logic with reusable Python modules and a Flask web interface for processing user requests.